FIITJEE
ALL INDIA TEST SERIES
JEE (Advanced)-2023
FULL TEST – I
PAPER –1
TEST DATE: 28-12-2022

ANSWERS, HINTS & SOLUTIONS

Physics

PART – I

Section – A

1. AD

2. ABC

3. ABCD

Sol. $\Delta E = \frac{hc}{\lambda}$

$\Delta E = \frac{hc}{\lambda(\text{in nm})}(\text{in eV})$

4. AD

5. BCD

6. BCD

7. C

8. A

9. B

10. C

Section – B

11. 250.00

12. 42.00

13. 2.96

Sol. Let b be the velocity and α the angle of projection for a fragment reaching the man. According to the questions the greatest value of u can be 20 m/s. If t be the time taken by this fragment to reach the man, then considering the horizontal and vertical motions of the fragment

$$8+\frac{1}{2}gt^2 = u\sin\alpha t$$

$$g^2t^4 - 4\left(u^2 - g\times 8\right)t^2 + 4\times 100 = 0$$

The period in which the man will be in danger on account of this fragment

$$= t_1 - t_2 = \sqrt{\left(t_1^2 + t_2^2\right) - 2t_1t_2} = \frac{2}{g}\left\{u^2 - 8g - 10g\right\}^{1/2}$$

14. 3.21

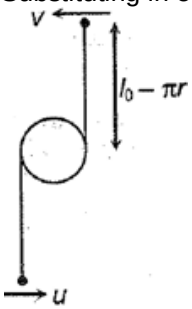
Sol. Using WET

$$\frac{1}{2}mv^2 - \frac{1}{2}mu^2$$

$$= -mg\left(l_0 + l_0 - \pi r\right) \text{ ..... (i)}$$

At highest point $v = \sqrt{\left(l_0 - \pi R\right)g}$

Substituting in equation (i)



$$v = \sqrt{5gl_0 - 4r\pi g} \cong 3.21m/\sec$$

15. 2.81

Sol. Velocity of watre coming out from hole

$$v = \sqrt{2gh} = \sqrt{2g1}$$

$$y = u_yt + \frac{1}{2}a_yt^2$$

$$1 = \sqrt{2g}t\sin 30° + \frac{1}{2}\times 10t^2$$

$$1 = \sqrt{2g}\,\frac{t}{2} + 5t^2$$

$$t = \frac{-2.24 \pm \sqrt{\left(2.24\right)^2 + 20}}{10} = 0.28\sec.$$

$$x_1 = v_xt = \sqrt{2g}\cos 30°\times 0.28 = 1.08m$$

$$x = x_1 + x_2 = 1.08 + 1.73 = 2.81m$$

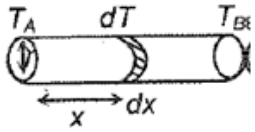
16. 500.00

Sol. Intensity of the source at the cross-section A

$$\Rightarrow l = \frac{P}{4\pi(50r)^2} = \frac{1.25\times10^3}{4\pi(50r)^2}$$

Power absorbed by the end A

$$T_B = \frac{0.003}{100000\times10^{-10}} = 300K$$



$$\int_0^j \frac{0.1T_A.dx}{4\times10^{-4}} = -\int_{T_A}^{T_B} TdT$$

$$\Rightarrow T_A = 500K$$

17. 40.00

Sol. When (1) and (2) are connected and (3) and (4) are open

$$P_1 = \frac{v^2}{R_1+R_2} \qquad .....(i)$$

When (3) and (4) are closed and (1) and (2) are connected

$$P_2 = \frac{v^2}{R_2+\left(\frac{R_2R_3}{R_2+R_3}\right)} \qquad .....(ii)$$

When (3) and (4) are connected and (1) and (2) are open

$$P_3 = \frac{v^2}{R_2+R_3} \qquad .....(iii)$$

When (3) and (4) are connected and (1) and (2) are closed

$$P_4 = \frac{v^2}{R+\left(\frac{R_1R_3}{R_1+R_3}\right)} \qquad .....(iv)$$

From (i) to (iv)

$$\frac{P_1}{P_3} = \frac{P_2}{P_4} \Rightarrow P_4 = 40W$$

18. 21.00

Sol. Accroding to Biot-Savert law,

$$\mathrm{B} = \frac{\mu_0}{4\pi}\frac{qv\sin\theta}{r^2}$$

$$\mathrm{B} = \frac{10^{-7}\times0.2\times4\times10^3\times3}{109\times\sqrt{109}} = 21\mu T$$

Chemistry

PART – II

Section – A

19. AD

Sol. Both Fe^{3+} & Mn^{2+} have 5 unpair electrons

Cu_2Cl_2 is colourless because Cu^+ has no unpair electrons

20. ABC

Sol. (A) The total lines obtained due to the splitting of a spectral line in the presence of magnetic effect is (2*l* + 1) as the presence of orbitals which have specific orientation in the presence of external field take up certain new orientation. The number of orbitals are equal to (2*l* + 1) and for each orbital one splitting takes place.

(B) As we move away from the nucleus the difference in the energy levels become lesser and lesser hence.

For 1st line of Lyman $\Delta E = E_1\left[\frac{1}{n_1^2}-\frac{1}{n_n^2}\right]$

$\Delta E_1 = E_1\left[\frac{3}{4}\right]$ – 1st line; $\Delta E_2 = E_1\left[\frac{8}{9}\right]$–2nd line

$\Delta E_3 = E_1\left[\frac{15}{16}\right]$–34rd line ; $\Delta E_4 = E_1\left[\frac{24}{25}\right]$ – 4th line;

$\Delta E_5 = E_1\left[\frac{35}{36}\right]$ – 5th line;

$\Delta E_2 - \Delta E_1 = E_1\left[\frac{8}{9}-\frac{3}{4}\right] = E_1\left[\frac{5}{36}\right]$

$\Delta E_3 - \Delta E_2 = E_1\left[\frac{15}{16}-\frac{8}{9}\right] =$

$E_1\left[\frac{135-128}{144}\right] = E_1\left[\frac{7}{144}\right]$

$\Delta E_4 - \Delta E_3 = E_1\left[\frac{24}{25}-\frac{15}{16}\right]$

$= E_1\left[\frac{16\times24-15\times25}{25\times16}\right] = E_1\left[\frac{9}{400}\right]$

We find that $(\Delta E_2 - \Delta E_1) > (\Delta E_3 - \Delta E_2) > (\Delta E_4 - \Delta E_3)$

Hence the distance also become lesser and lesser.

(C) The distance from the nucleus for maximum probability of finding electrons is 0.53 Å. This is not on the circumference of the orbital but is in the vicinity of the circumference.

21. ABC

Sol. Because of the H–bonding NH_4^+ is more extensively solvated than Na^+, hence NH_4^+ salts is more soluble than Na^+ salts.

The polarising power of Sn^{4+} is greater than Sn^{2+} , hence $SnCl_4$ is more covalent

The polarisibility of Cl^- is greater than F^- because of greater size.

Formal charge

$$= \begin{pmatrix} \text{no. of valence} \\ \text{electrons of the} \\ \text{atom} \end{pmatrix} - \begin{pmatrix} \text{no. of} \\ \text{nonbonded} \\ \text{electrons} \end{pmatrix} - \frac{1}{2}\begin{pmatrix} \text{no. of} \\ \text{bonded} \\ \text{electrons} \end{pmatrix}$$

In SO_2 , formal charge of S = $6 - 2 - \frac{1}{2} \times 4 = 2$

22. AB

23. ABCD

24. AC

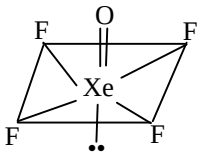
Sol. $Hg_2^{2+} + 2I^- \rightarrow Hg_2I_2$(green ppt) ;
$Hg_2^{2+} + H_2S \rightarrow$ no ppt

25. B

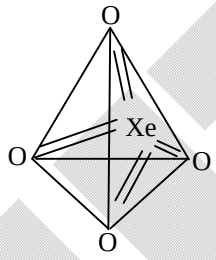
26. A

Sol.

(I) $XeOF_4$:

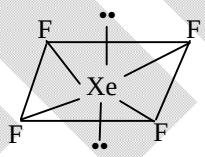


one ℓ p of electrons ; sp^3d^2 hybridisation ; square pyramidal
(I) $\rightarrow$ (Q,S)
(II) XeO_4 : tetrahedral and no ℓ p of electrons



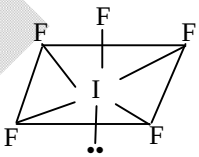
(II) $\rightarrow$ (P, S)

(III) XeF_4 :



sp^3d^2 hybridisation ; planar ; two ℓ ps of electrons
(III) $\rightarrow$ R

(IV) IF_5 :



sp^3d^2 hybridisation ; one ℓ p. of electrons
(IV) $\rightarrow$ (Q,S)

27. D

28. C

Section – B

29. 0.90

Sol. Let no. of moles of CH_4 present = n_1 mol.

Let no. of moles of C_2H_2 present = n_2 mol.

$CH_4 + 2O_2 \longrightarrow CO_2 + 2H_2O$

$n_1 \qquad\qquad n_1$

$C_2H_2 + \frac{5}{2} O_2 \longrightarrow 2CO_2 + H_2O$

$n_2 \qquad\qquad 2n_2$

Total no. of moles at initial = $n_1 + n_2$

Total no. of moles at final = $n_1 + 2n_2$

At constant V & T,

$$\frac{P_1}{P_2} = \frac{\text{No. of moles at initial}}{\text{No. of moles at final}}$$

$$\frac{63}{69} = \frac{n_1+n_2}{n_1+2n_2} \Rightarrow \frac{21}{23} = \frac{n_1+n_2}{n_1+2n_2}$$

$$\Rightarrow \frac{n_2}{n_1} = \frac{2}{19} \Rightarrow \frac{n_1+n_2}{n_1} = \frac{21}{19}$$

$$\therefore \frac{n_1}{n_1+n_2} = \frac{19}{21}$$

30. 0.39

Sol. $\Delta u = Q + W = 0 + \int -P_{ext.}dV = -P_{ext}(V_2 - V_1)$

$$\Rightarrow C_V (T_2 - T_1) = -P_{ext.}\left(\frac{RT_2}{P_2} - \frac{RT_1}{P_1}\right);$$

Here $P_{ext} = P_2$

$$\Rightarrow \frac{5}{2}R(673-373) = -673R + \left(\frac{373R}{0.1013}\right)P_2$$

$\Rightarrow P_2 = 0.3865$ M Pa

31. 800.00

Sol. $I_2 + I^- \underset{}{\overset{KC}{\rightleftharpoons}} I_3^-$

$51.25 \times 10^{-3} \qquad 0.1$

At Equilibrium $\quad 1.25 \times 10^{-3}$(M) $\quad 50 \times 10^{-3}$(M) $\quad 50 \times 10^{-3}$(M)

$$K_C = \frac{50\times10^{-3}}{50\times10^{-3}\times1.25\times10^{-3}} = \frac{1000}{1.25} = 800$$

32. 0.40

Sol. $N_2O_5(g) \rightleftharpoons 2NO_2(g) + \frac{1}{2}O_2(g)$

Initially : 600 torr — —

At equilibrium: (600 – p) torr 2p torr $\frac{p}{2}$ torr

Total press is $\left(600+\frac{3p}{2}\right)$ torr

$\therefore$ $600 + \frac{3p}{2} = 960$

or $\frac{3p}{2} = 360$

or $p = 240$ torr

$\therefore$ fraction of N_2O_5 (g) decomposed = $\frac{240}{600} = 0.4$

33. 4.00

Sol. $s = \frac{10^3 \times \kappa}{\lambda_m^{\infty}(AgCl)} = \frac{10^3 \times 2.6 \times 10^{-6}}{(63+67)}$

$= 2 \times 10^{-5}$ (M)

$\therefore$ $K_{sp} = (2 \times 10^{-5})^2 = 4 \times 10^{-10}$.

34. 0.50

35. 64.36

Sol. Vol. of all atoms in 1.68 gm

argon $= \frac{1.68}{40} \times N_A \times \frac{4}{3} \times \pi \times (1.5 \times 10^{-8})^3 = 0.3564$

vol. of solid argon = 1 cm^3

% empty space = (1 – 0.3564) × 100 = 64.36

36. 1000.00

Sol. Initially $P_m = P^o_X.X_X + P^o_Y.X_Y$

$550 = P^o_X\left(\frac{1}{1+3}\right) + P^o_Y\left(\frac{3}{1+3}\right)$

or $P°_X + 3P°_Y = 2200$ ...(i)

When 1 mole of Y is further added to it

$P_m = P^o_X.X_X + P^o_Y.X_Y$

$560 = P^o_X\left(\frac{1}{1+4}\right) + P^o_Y\left(\frac{4}{1+4}\right)$

$\therefore$ $P°_X + 4P°_Y = 2800$ ...(ii)

By eqs. (i) and (ii)

P^o_X = 400 mm and $P°_Y$ = 600 mm.

Mathematics

PART – III

Section – A

37. ACD

Sol. $\because (A+B)(A-B)=(A-B)(A+B)$

Now $C=\left(A^T+B^T\right)(A+B)(A-B)^{-1}$

$=(A-B)(A+B)(A-B)^{-1}$

$=(A+B)(A-B)(A-B)^{-1}$

$=A+B$

Now $C^T=A^T+B^T=A-B$

$C+C^T=2A$

38. BD

Sol. Let $t_r=\left(\frac{1}{r(r+1)}\right)^3=\left(\frac{1}{r}-\frac{1}{r+1}\right)^3=\frac{1}{r^3}-\frac{1}{(r+1)^3}-\frac{3}{r(r+1)}\left(\frac{1}{r(r+1)}\right)$

$=\frac{1}{r^3}-\frac{1}{(r+1)^3}-3\left(\frac{(r+1)-r}{r(r+1)}\right)^2$

$t_r=\frac{1}{r^3}-\frac{1}{(r+1)^3}-3\left(\frac{1}{r^2}+\frac{1}{(r+1)^2}\right)+6\left(\frac{1}{r}-\frac{1}{r+1}\right)$

$\sum_{r=1}^{n}t_r=\left(1-\frac{1}{(n+1)^3}\right)-3\left(\frac{\pi^2}{6}+\frac{\pi^2}{6}-1\right)+6\left(1-\frac{1}{n+1}\right)$

$\lim_{x\to\infty}\sum t_r=10-\pi^2$

39. BCD

Sol. $f(x)=x^2+\int_0^x e^{-t}f(x-t)\,dt$

$=x^2+\int_0^x e^{-(x-t)}.f\left(x(x-t)\right)dt$

$=x^2+e^{-x}.\int_0^x e^t.f(t)\,dt$

$\therefore e^x.f(x)=x^2.e^x+\int_0^x e^t.f(t)\,dt$

Differentiate with respect to x, we get $f'(x)=x^2+2x$

$\Rightarrow f(x)=\frac{x^3}{3}+x^2+x, also\ f(0)=0$

$\Rightarrow c=0$

$\Rightarrow f(x)=\frac{x^3}{3}+x^2$

40. ABC

Sol. When 9 is not included $=\frac{{}^8C_3.{}^8C_3-{}^8C_3}{2}$

and required probability is $=\frac{1}{3}+=\frac{{}^8C_3.{}^8C_3-{}^8C_3}{2.{}^9C_3.{}^8C_3}=\frac{37}{56}$

41. AB

Sol. By property: $PQ\cdot P'Q=b^2$ and area of $\Delta CLG=ab$

42. ABC

Sol. $0<x<\frac{\pi}{2}\Rightarrow\frac{\sin x}{x}$ is decreasing and $\sin x<x<\tan x$

$\Rightarrow\frac{\sin(\sin x)}{\sin x}>\frac{\sin x}{x}>\frac{\sin(\tan x)}{\tan x}$

$\Rightarrow I_1>I_2>I_3$

43. A

44. B

Sol. (I) clearly, number of solutions is 2.

(II) $f(x)=\cos^{-1}\left(\frac{x^2}{1+x^2}\right)=\cos^{-1}\left(1-\frac{1}{1+x^2}\right)$

$t=1-\frac{1}{1+x^2}\Rightarrow\ t\in[0,1)$

$\therefore\cos^{-1}t\in\left(0,\frac{\pi}{2}\right]=(a,b]\Rightarrow[a+b]=1$

(III) $12\left(\tan^2\left(\tan^{-1}\alpha\right)+\tan^2\left(\tan^{-1}\beta\right)\right)=12\left(\alpha^2+\beta^2\right)$

$=12\left((\alpha+\beta)^2-2\alpha\beta\right)$

$=12\left(\frac{9}{4}-2.1\right)=3$

(IV) $\sin\left(\cos^{-1}x\right)=\sin\left(\frac{\pi}{2}-\sin^{-1}x\right)=\cos\left(\sin^{-1}x\right)=t$

$f(x)=\sin\left(\cos^{-1}t+\sin^{-1}t\right)=\sin\frac{\pi}{2}=1$

$$\sum_{x=1}^{4} f\left(\frac{3x}{16}\right)=1+1+1+1=4$$

45. C

Sol. (a) $3x+y-z=0$ .....(1)

$$x-\frac{py}{4}+z=0 \text{ .....}(2)$$

$$2x-y+2z=q \text{ ....}(3)$$

Eqn. (2) × (2) – eqn. (3)

$$\Rightarrow\left(1-\frac{p}{2}\right)y=4-q$$

For unique solution, $p\neq 2,\ q\in N \Rightarrow$ number of ordered pairs (p,q) in $[1,10]$ are 90.

For infinite solution, $p=2$ and $q=4 \Rightarrow$ exactly one ordered pair

For no solution, $p=2$ and $q\neq 4 \Rightarrow$ number of ordered pairs (p,q) in $[1,10]$ are 9

46. D

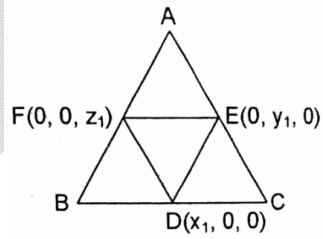
Section – B

47. 8.00

Sol. Let the intercept of plane on the axes be $(x_1,0,0),(0,y_1,0),(0,0,z_1)$

So, equation is $\frac{x}{x_1}+\frac{y}{y_1}+\frac{z}{z_1}=1$

$\because EF \parallel BC$ and $EF=\frac{1}{2}BC=4\left(y_1^2+z_1^2\right)=a^2$



Similarly $4\left(x_1^2+y_1^2\right)=c^2$ and $4\left(z_1^2+x_1^2\right)=b^2$

$$\Rightarrow a^2+b^2+c^2=8\left(x_1^2+y_1^2+z_1^2\right)$$

$$\Rightarrow k=8$$

48. 833.33

Sol. $\because \int_a^x e^{f(t)}dt=\int_0^x g(x-t)dt+2x+3$

$\Rightarrow \int_a^x e^{f(t)}dt=\int_0^x g(t)dt+2x+3$ (Using King Property)

Differential both sides, we get

$e^{f(x)}=g(x)+2$

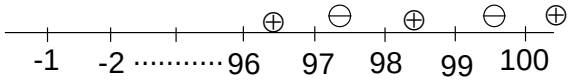
$\Rightarrow g(x)=e^{f(x)}-2$

$\Rightarrow g'(x)=e^{f(x)}.f'(x)$

$\because e^{f(x)}$ is always greater than zero.

$\therefore$ Sign of g'(x) is same as sign of $f'(x)$.

$\therefore$ Sign of g'(x)



Clearly, local extreme (maximum or minimum) will occur at x=99,97,95,….,3,1

$\therefore$ Sum of all the values=1+3+5+…+99= $\frac{50}{2}[2\times1+(50-1)\times2]=2500$

49. 5.00

Sol. $\lim\limits_{y\to1}\frac{100y^{100}}{y^{100}-1}-\frac{50.y^{50}}{y^{50}-1}$

$\lim\limits_{y\to1}100-50-\left(\frac{100}{1-y^{100}}-\frac{50}{1-y^{50}}\right)$

L=50-L $\therefore$ L= $\frac{50}{2}=25$

50. 20.00

Sol. $S_i=\frac{20}{2}\left(2+19di\right)$

$$\frac{2S_3S_1-\left(S_1+S_3\right)}{S_1+S_3-2S_2}=\frac{200\left(2+19d_1\right)\left(2+19d_3\right)-100\left(2+19d_2\right)\left(4+19\left(d_1+d_3\right)\right)}{10\left(4+19\left(d_1+d_3\right)-4-38d_2\right)}$$

$$=10\left(\frac{8+76\left(d_1+d_3\right)+722d_1d_3-8-38\left(d_1+d_3+2d_2\right)-361d_2\left(d_1+d_3\right)}{19\left(d_1+d_3-2d_2\right)}\right)$$

$$=10\left(\frac{38\left(d_1+d_3-2d_2\right)+361\left(2d_1d_3-d_2\,d_1+d_3\right)}{19\left(d_1+d_3-2d_2\right)}\right)$$

$$=20$$

51. 10.00

52. 25.00

Sol. $\left(\sin x+\cos x\right)\left(\tan^2 x-\left(1+\sqrt{3}\right)\tan x+3+\sqrt{3}\right)=3\left|\sin x+\cos x\right|$

Case – 1: $\sin x+\cos x\geq0$

$$\left(\sin x+\cos x\right)\left(\tan^2 x-\left(1+\sqrt{3}\right)\tan x+\sqrt{3}\right)=0$$

$$\tan x=-1,\ 1,\ \sqrt{3}$$

$$\Rightarrow\qquad x=\frac{3\pi}{4},\frac{\pi}{4},\frac{\pi}{3},\frac{7\pi}{4}$$

Case- 2 : $\sin x+\cos x\leq0$

$$\tan x=-1$$

$\therefore \text{Sum} = \dfrac{37\pi}{12}$

53. 151.00

Sol. $S = 2 + 3\cos x + 4\cos^2 x + \ldots\ldots\ldots\ldots\infty$

$$\frac{S.\cos x = \quad 2\cos x + 3\cos^2 x + \ldots\ldots\ldots\ldots\infty}{S(1-\cos x) = 2 + \cos x + \cos^2 x \ldots\ldots\ldots\ldots\infty}$$

$$= 1 + \frac{1}{1-\cos x}$$

$$S = \frac{1}{1-\cos x} + \frac{1}{(1-\cos x)^2}$$

$|5\cos x + 4| + |5\cos x - 2| = 6$

$\Rightarrow \cos x \in \left[-\dfrac{4}{5}, \dfrac{2}{5}\right]$

For the least value of S, $\cos x = -4/5$

$\therefore a + b = 151$

54. 14.00

Sol. $\left(2 + x^3 (3+x)^2\right)^{10}$

$\downarrow$

${}^{10}C_r . 2^{10-r} . x^{3r} (3+x)^{2r}$

$\downarrow$

${}^{2r}C_k . 3^{2r-k} . x^k$

$3r + k = 8 \Rightarrow k = 2,\ r = 2$

$\therefore$ Coefficient of $x^8 = {}^{10}C_2 . 2^8 . {}^4C_2 . 3^2 = 5.2^9 . 3^5$